

Matthew R. Conrad

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Skills

Languages: Python, C++, Java, Swift, TypeScript, JavaScript, SQL, Bash

Frameworks & Tools: SwiftUI, HealthKit, AVFoundation, AlarmKit, WidgetKit, StoreKit 2, CoreML, Vision, PyTorch, TensorFlow, React, Pandas, NumPy, Scikit-learn, Smithy, Coral, Dagger, Lombok, Docker

Cloud & Infrastructure: AWS (CDK, IAM, SQS, CloudTrail, DynamoDB, Bedrock), Firebase (Firestore, Cloud Functions, Auth, FCM), Postgres

Techniques: Reinforcement Learning, Ensemble Methods, Time Series Analysis, Explainability (SHAP/GAM), Clustering, Statistical Modeling, Kernel Methods

Security: Multi-Party Computation (MPC, Fireblocks), Smart Contract Custody, Cyber Threat Analysis, Incident Response, Zero-Downtime Patching

Education

Georgia Institute of Technology *In Progress*
Master of Science in Computer Science (Machine Learning)

University of Michigan *Aug 2021 – May 2024*
Bachelor of Science in Computer Science

Selected Publications

- Harmanpreet Kaur, **Matthew R. Conrad**, et al. 2024. "Interpretability Gone Bad: The Role of Bounded Rationality in How Practitioners Understand Machine Learning." [Proc. ACM HCI, 2024](#) [🔗](#).

Experience

Software Development Engineer II *Herndon, VA*
Amazon Web Services *Nov 2025 – Present*

- Lead engineer for SDK distribution on a greenfield AWS product launched to external customers in June 2026; designed, implemented, tested, and deployed the end-to-end delivery path
- Productized Smithy-generated SDKs for Java, Go, Rust, and Python plus AWS CLI integration, backed by Java-based authentication and artifact-retrieval APIs; defined packages, access controls, and integration patterns for external customers and partner teams
- Codified a multi-account, multi-region artifact platform in AWS CDK, provisioning S3 buckets, deployment pipelines, IAM roles and permissions, alarms, canaries, integration-test infrastructure, and deployment configurations
- Shipped cross-team performance metrics request/approval feature spanning data model, API, and backend
- Engineered secure SDK delivery with IAM allowlisting, cross-account role assumption and presigned retrieval; added CloudTrail auditing and partition-safe SQS handling for compliance across AWS environments
- Won first place at an AWS AI hackathon for an on-call alarm agent that accelerates incident triage and resolution

Software Engineer, Web3 Trading Infrastructure *New York, NY, Remote*
Concord Systems Corp. [🔗](#) *Jun 2025 – Aug 2025*

- Engineered Fireblocks MPC-based transaction approval into production OTC trading systems
- Designed flexible trade execution logic supporting both atomic and non-atomic trade flows
- Built and deployed backend services for transaction signing, order routing, and database migrations
- Strengthened security and consistency across trading flows through multi-party coordination protocols

Principal Computer Systems Architect *Los Angeles, CA*
Northrop Grumman *May 2023 – Oct 2025*

- Served as a lead network engineer for a large-scale, high-availability datacenter network spanning 100+ nodes, owning Cisco Layer 2 Ethernet switching and Layer 3 TCP/IP routing for mission-critical real-time systems
- Configured and optimized dynamic routing with BGP, OSPF, and RIP; designed redundant paths and tuned route convergence and failover for high availability under sub-millisecond latency requirements
- Implemented ACLs, QoS policies, network segmentation, and Palo Alto firewall controls to secure distributed systems and prioritize latency-sensitive data and VoIP traffic
- Diagnosed latency, packet loss, and throughput using SNMP monitoring, Wireshark, and tcpdump; performed NIC/kernel tuning, routing-table analysis, and cross-layer fault isolation across containers and applications
- Led compliance hardening and rolling, zero-downtime patch cycles across the 100+ node environment while preserving availability of mission-critical services

- Built automated MySQL and MS Access inventory systems tracking more than \$1M in datacenter infrastructure assets

Machine Learning Researcher

University of Michigan

Ann Arbor, MI
Oct 2021 – Oct 2025

- Conducted deep learning research applying convolutional neural networks (CNNs) using Keras to study model trust, interpretability, and failure modes.
- Co-authored CSCW 2024 and CHI 2026 papers on cognitive biases and contrastive explanations in AI model evaluation.
- Developed and evaluated a quiz-based active learning framework to improve user understanding of AI limitations via contrastive example selection.
- Designed end-to-end pipelines for automated selection of model failure cases using statistical similarity (SIFT, clustering) and human-meaningful contrastive pairs.
- Applied SHAP, GAMs, and regression analysis to connect model predictions with behavioral outcomes in user studies (N=119).
- Led experimental design, IRB approval, and full deployment of a cross-sectional study measuring AI literacy outcomes.

Information Technology Analyst

Seifert Technologies Inc.

Massillon, OH
Feb 2021 – Jun 2021

- Built Power BI dashboards (DAX) for analytics across logistics, billing, and production cycles.
- Supported ERP systems, Microsoft 365, and remote IT infrastructure in a client-facing support role.
- Automated backup verification and streamlined invoicing workflows via QuickBooks APIs.

Counterintelligence Agent, Team Chief, Cyber Network Operator

United States Marine Corps

Various Locations
Aug 2012 – Jan 2021

- Directed cyber threat assessments and incident response protocols for sensitive operations, including the F-35 program.
- Conducted HUMINT and SIGINT operations; contributed to secure information systems across Indo-Pacific deployments.
- Designed remote operational networks, integrating encryption and secure routing for joint-force communication.
- Led 40+ briefings on information security and network defense; trained 750+ personnel in threat awareness.
- Managed SIPRNet services for 1,500+ users, resolving infrastructure and access control issues in high-security zones.

Projects

Deva

Python, FastAPI, OpenGrep,
Docker, GCP

- Architected and built a local-first AI security platform with a shared scanning engine and authenticated API serving IDE, CLI, CI, extension, and MCP clients; integrated 415-rule SAST, dependency/SBOM analysis, compliance mapping, and automated remediation
- Engineered a secure cross-platform software supply chain with hash-locked dependencies, immutable artifacts, Ed25519-signed manifests and entitlements, isolated signing roles, integrity verification, rollback, and automated release smoke tests
- Built isolated Cloud Run control-plane services with least-privilege IAM, Secret Manager-backed credentials, short-lived capability grants, offline-verifiable authorization, and managed, local, or BYOK model routing without exposing provider secrets to clients

Zenblis

TypeScript, React, Firebase,
Stripe, Google Maps

- Built and launched a two-sided senior-living marketplace combining location-based facility discovery with a multi-tenant SaaS platform for community operators
- Engineered lead capture, CRM pipelines, email automation, analytics, listing management, subscriptions, and Google Calendar integration with Firebase Functions, Firestore, Stripe, Postmark, and Google APIs
- Developed a role-based admin platform and block-based CMS with scheduled publishing, revision history, structured SEO data, server-rendered metadata, automated sitemaps, and Playwright end-to-end coverage

Yammoing

Swift, SwiftUI, TypeScript,
Firebase, Gemini

- Architected and shipped a full-stack iOS health and nutrition platform integrating barcode scanning, multimodal AI analysis, HealthKit, and StoreKit 2 subscriptions
- Secured user data and billable APIs with authenticated Cloud Functions, App Attest/DeviceCheck, Secret Manager, least-privilege Firestore rules, server-controlled entitlements, and atomic tier-based rate limiting
- Built resilient integrations across Gemini, USDA, OpenFoodFacts, and Foursquare with centralized authorization, input validation, quota enforcement, and model fallback handling

Certifications

- CompTIA Security+ (exp. 9/2028)
- CompTIA A+ (exp. 9/2028)

Other

- Security Clearance: TS/SCI with CI-scope polygraph
- Awards: Navy & Marine Corps Commendation Medal (2020), Navy & Marine Corps Achievement Medal (2016, 2018)